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United States Patent	12384113
Kind Code	B2
Date of Patent	August 12, 2025
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Filament replacement mechanism for 3D printer and 3D printing system

Abstract

Various embodiments provide a filament replacement mechanism for a 3D printer and a 3D printing system. In those embodiments, the filament replacement mechanism includes: a support; a filament guide unit disposed on the support and including a plurality of secondary filament guide tubes and a filament guide opening connecting with the plurality of secondary filament guide tubes, where the filament guide opening is configured to be jointed with a main filament guide tube of the 3D printer; and a plurality of filament replacement units disposed on the support.

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Appl. No.:	18/460596
Filed:	September 04, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230405933 A1	Dec. 21, 2023

Foreign Application Priority Data

CN	202210058867.7	Jan. 17, 2022
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Related U.S. Application Data

continuation parent-doc WO PCT/CN2023/071921 20230112 PENDING child-doc US 18460596

Publication Classification

Int. Cl.: B29C64/321 (20170101); B65H51/12 (20060101); B33Y30/00 (20150101)

U.S. Cl.:

CPC B29C64/321 (20170801); B65H51/12 (20130101); B33Y30/00 (20141201); B65H2404/421 (20130101)

Field of Classification Search

CPC: B29C (64/321); B33Y (10/00); B33Y (30/00); B65H (2404/421)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Patent Application No. PCT/CN2023/071921, filed on Jan. 12, 2023, which claims priority to Chinese Patent Application No. 202210058867.7, filed on Jan. 17, 2022. The disclosures of the aforementioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

(1) The present disclosure relates to the field of three-dimensional (3D) printing technology, and in particular to a filament replacement mechanism for a 3D printer and a 3D printing system.

BACKGROUND

(2) 3D printer, also known as three-dimensional printer, constructs a 3D object by layer-by-layer printing. 3D printer includes a printing head for extruding a printing material and a print table for depositing the printing material to form a 3D object. The printing head is configured to move relative to the print table and to extrude the printing material onto the print table while moving. The printing material is deposited layer by layer on the surface of the print table and fused together to form the 3D object.

(3) With the rapid development of 3D printing technology, there are increasingly diverse demands for colors and materials of printed objects. How to satisfy the diversified printing demands is gradually turning into a research hotspot.

SUMMARY

(4) It would be advantageous to provide a mechanism that alleviates, mitigates or even eliminates one or more of the above-mentioned problems.

(5) According to an aspect of the present disclosure, a filament replacement mechanism for a 3D printer is provided, including: a support; a filament guide unit disposed on the support and including a plurality of secondary filament guide tubes and a filament guide opening connecting with the plurality of secondary filament guide tubes, where the filament guide opening is configured to be jointed with a main filament guide tube of the 3D printer; and a plurality of filament replacement units disposed on the support, each of the plurality of filament replacement units automatically performing a loading operation and an unloading operation on a corresponding filament spool of a plurality of filament spools for the 3D printer. The filament replacement mechanism for a 3D printer according to the present disclosure has the beneficial effects that a variety of filaments can be automatically provided for the 3D printer according to printing needs, and the replacement of filament spools is also facilitated. Each of the filament replacement units includes: a supporting shaft assembly rotatably connected to the support and configured to be in contact with a rim of a corresponding filament spool of the plurality of filament spools so as to support the corresponding filament spool; and a loading and unloading mechanism configured to drive a filament wound around the corresponding filament spool to enable the filament to be guided

into the main filament guide tube via one corresponding secondary filament guide tube of the plurality of secondary filament guide tubes when the 3D printer performs a loading operation, and configured to drive the corresponding filament spool to rotate such that the filament withdrawn from the main filament guide tube via the filament guide unit is rewound around the corresponding filament spool when the 3D printer performs an unloading operation.

(6) According to an aspect of the present disclosure, a 3D printing system is provided, comprising a 3D printer and a filament replacement mechanism according to the foregoing aspect.

(7) These and other aspects of the present disclosure will become clear on the basis of the embodiments described hereinafter, and will be elucidated with reference to the embodiments described hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Further details, features and advantages of the present disclosure are disclosed in the following description of exemplary embodiments in conjunction with the accompanying drawings. In the accompanying drawings:

(2) FIG. 1 shows a stereoscopic structure diagram of a filament replacement mechanism according to some exemplary embodiments of the present disclosure.

(3) FIG. 2 shows a side view of a filament replacement mechanism according to some exemplary embodiments of the present disclosure.

(4) FIG. 3 shows a rear view of a filament replacement mechanism according to some exemplary embodiments of the present disclosure.

(5) FIG. 4 shows a stereoscopic structure diagram of a filament replacement mechanism after the removal of a filament spool and a loading and unloading mechanism according to some exemplary embodiments of the present disclosure.

(6) FIG. 5 shows a stereoscopic structure diagram of a filament replacement mechanism after the removal of a filament spool and a top cover according to some exemplary embodiments of the present disclosure.

(7) FIG. 6 shows a stereoscopic structure diagram of a filament replacement mechanism according to some exemplary embodiments of the present disclosure.

DETAILED DESCRIPTION

(8) It will be understood that, although the terms “first,” “second,” “third”, etc., may be used herein to describe various elements, components, regions, layers and/or portions, such elements, components, regions, layers and/or portions shall not be limited by these terms. These terms are merely intended to distinguish one element, component, region, layer or portion from another element, component, region, layer or portion. Thus, the first element, component, region, layer or portion discussed below may be referred to as a second element, component, region, layer or portion without departing from the teachings of the present disclosure.

(9) For ease of description, spatially relative terms, such as “under”, “beneath”, “lower”, “below”, “on top of”, and “upper” and the like may be used herein to describe the relationship between one element or feature and another element (s) or feature (s) shown in the figures. It will be understood that, these spatially relative terms are intended to encompass different orientations of a device in use or operation in addition to the orientation depicted in the figure. For example, if the device in the figure is turned over, an element described as “beneath another element or feature” or “under another element or feature” or “below another element or feature” would then be oriented as “on top of another element or feature”. Thus, exemplary terms “beneath” and “below” may encompass both orientations “on top of” and “beneath”. Similarly, terms such as “before” or “prior to” and “after” or “next” may be used, for example, to indicate the order in which light passes through

elements. The device can be otherwise oriented (rotated 90 degrees or at other orientations), and the spatially relative descriptors used herein are construed accordingly.

(10) The terms used herein are merely for the purpose of describing specific embodiments, and are not intended to limit the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that, the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of described features, entirities, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, entirities, steps, operations, elements, components, and/or groups thereof. As used herein, the term “and/or” includes any of and all combinations of one or more of the associated listed items, and the phrase “at least one of A and B” means only A, only B, or both A and B.

(11) It will be understood that, when an element or layer is referred to as being “on another element or layer,” “connected to another element or layer,” “coupled to another element or layer” or “adjacent to another element or layer”, it can be directly on, directly connected to, directly coupled to or directly adjacent to another element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on another element or layer”, “directly connected to another element or layer”, “directly coupled to another element or layer”, or “directly adjacent to another element or layer”, there are no intervening elements or layers present. However, in no case should “on” or “directly on” be construed as requiring one layer to completely cover an underlying layer.

(12) Unless otherwise defined, all terms including technical terms and scientific terms used herein have the same meanings as those generally understood by those of ordinary skill in the art to which the present disclosure belongs. It will be further understood that, terms, such as those defined in commonly used dictionaries, should be construed as having meanings consistent with meaning of these terms in the relevant art and/or in the context of this specification, and should not be construed in an idealized or overly formal sense, unless explicitly defined herein.

(13) In the related art, it is required to manually replace filament spools when a 3D printer prints different colors or different materials, which results in complicated operation and low efficiency.

(14) In order to alleviate, mitigate or even eliminate one or more of the above-mentioned problems, the embodiments of the present disclosure provide a filament replacement mechanism for a 3D printer and a 3D printing system, such that various filaments can be automatically provided for the 3D printer according to printing needs, and the replacement of filament spools is also facilitated.

(15) The embodiments of the present disclosure are described in detail below with reference to the accompanying drawings.

(16) FIG. 1 shows a stereoscopic structure diagram of a filament replacement mechanism according to some embodiments of the present disclosure. FIG. 2 shows a side view of a filament replacement mechanism according to some embodiments of the present disclosure. FIG. 3 shows a rear view of a filament replacement mechanism according to some embodiments of the present disclosure.

(17) Referring to FIG. 1 to FIG. 3, some embodiments of the present disclosure provide a filament replacement mechanism for a 3D printer, including: a support **100**, a plurality of filament replacement units **300** and a filament guide unit **200**.

(18) The filament guide unit **200** is disposed on the support **100**. The filament guide unit **200** includes a plurality of secondary filament guide tubes **210** and a filament guide opening **220** connecting with the plurality of secondary filament guide tubes **210**. The filament guide opening **220** is configured to be jointed with a main filament guide tube of the 3D printer.

(19) The plurality of filament replacement units **300** are disposed on the support **100**. Each of the plurality of filament replacement units **300** automatically performs a loading operation and an unloading operation on a corresponding filament spool of a plurality of filament spools for the 3D printer, where each filament replacement unit **300** includes: a supporting shaft assembly **310** and a loading and unloading mechanism **320**. The supporting shaft assembly **310** is rotatably connected

to the support **100** and may be configured to be in contact with a rim **410** of a corresponding filament spool **400** of the plurality of filament spools **400** to support the corresponding filament spool **400**. The loading and unloading mechanism **320** is configured to drive a filament wound around the corresponding filament spool **400** to enable the filament to be guided into the main filament guide tube via one corresponding secondary filament guide tube **210** of the plurality of secondary filament guide tubes **210** when the 3D printer performs a loading operation, and is configured to drive the corresponding filament spool **400** to rotate such that the filament withdrawn from the main filament guide tube via the filament guide unit **200** is rewound around the corresponding filament spool **400** when the 3D printer performs an unloading operation.

(20) In some embodiments of the present disclosure, the support **100** may be made of a plastic material or a metal material, and the support **100** is configured to mount the filament guide unit **200** and the filament replacement unit **300**.

(21) In an embodiment of the present disclosure, the filament guide unit **200** includes a plurality of secondary filament guide tubes **210**, and the secondary filament guide tubes **210** are all connected to one filament guide opening **220**. In some embodiments, the plurality of secondary filament guide tubes **210** may be manufactured by an integrated molding process, but the present disclosure is not limited thereto.

(22) In some embodiments, the filament guide unit **200** may further include a multi-channel member **240**, and the multi-channel member **240** connects the plurality of secondary filament guide tubes **210** to the filament guide opening **220**. The multi-channel member **240** may have an outlet configured to be jointed with the main filament guide tube, and a plurality of inlets connecting with the outlet, where the inlets may be configured to connect one secondary filament guide tube **210**, and the outlet may constitute the filament guide opening **220**. In some embodiments, an auxiliary extrusion wheel (not shown) is disposed inside the multi-channel member **240** to assist the loading and unloading mechanism **320** in performing a loading operation and an unloading operation. In some embodiments, a filament sensor (not shown) may be further disposed inside the multi-channel member **240** to detect the position of the filament in the filament guide tube.

(23) Each filament spool **400** may include two baffles spaced along an axis of rotation of the filament spool and a cylindrical or cylinder-like winding portion for winding of the filament located in between the two baffles, where the radius of the baffle is greater than the radius of the winding portion so as to prevent the filament from falling off the winding portion. The cylindrical surface of each baffle may constitute the rim **410** of the filament spool **400**. The free end of the filament of each filament spool **400** may stretch into a corresponding secondary filament guide tube **210**.

(24) The filament replacement unit **300** includes a supporting shaft assembly **310** and a loading and unloading mechanism **320**. The supporting shaft assembly **310** can be rotatably connected to the support **100**. For example, the supporting shaft assembly **310** may be connected to the support **100** via a bearing. The cylindrical surface of the supporting shaft assembly **310** is configured to be in contact with the rim **410** of the filament spool **400** to support the filament spool **400**. The supporting shaft assembly **310** may be in force transfer contact with the rim **410** of the filament spool **400** such that when one rotates, the other can follow the former to rotate. The supporting shaft assembly **310** may be of various structures, which will be further described later.

(25) In some embodiments, the loading and unloading mechanism **320** may be configured to drive the filament spool to rotate and achieve loading and unloading. For example, the loading and unloading mechanism **320** may include a loading motor and an unloading motor which may be separately in transmission connection with the filament spool. Taking the loading motor as an example, the loading motor can drive a friction wheel that is in contact with the rim **410** of the filament spool **400** to achieve rotation of the filament spool. For another example, the loading motor can also directly drive the filament to move by driving a friction wheel in direct contact with the filament, and drive the filament spool to rotate by the movement of the filament.

(26) Either way, when the 3D printer performs a loading operation, the loading motor may be configured to drive the filament spools **400** to rotate in a forward direction, such that the filament wound around the corresponding filament spool **400** is guided into the main filament guide tube via one corresponding secondary filament guide tube **210** and after penetrating through the filament guide opening **220**. When the 3D printer performs an unloading operation, the unloading motor may be configured to drive the corresponding filament spool **400** to rotate in a reverse direction, such that a free end of the filament is withdrawn from the main filament guide tube via the filament guide unit **200**. The withdrawn filament can be rewound around the corresponding filament spool **400**, thereby avoiding the problem that the filament is accumulated around the filament spool **400** and wound around other mechanisms after return of materials.

(27) When it is required to replace the current filament, the filament can be withdrawn from the main filament guide tube through the loading and unloading mechanism **320** corresponding to the current filament, and then a required filament is guided into the main filament guide tube through the loading and unloading mechanism **320** corresponding to the required filament. During printing, the printing motor of the 3D printer can drive the filament in the main filament guide tube to move, so as to drag the filament spool **400** to rotate. At this time, the loading motor and the unloading motor can be in drive disengagement from the filament spool **400**, such that the movement of the filament spool **400** is not influenced.

(28) In addition, the loading and unloading mechanism **320** can be arranged to a structure having only one driving motor except for the loading motor and the unloading motor, and the forward and reverse rotation of the filament spool can be realized through the forward and reverse rotation of the driving motor.

(29) Through the arrangement of the filament guide unit **200** and the filament replacement unit **300**, a required filament can be automatically provided for the 3D printer to achieve printing of three-dimensional objects of various colors or various materials, and the unnecessary filament after the return of materials can be automatically wound around the filament spool **400** to keep a clean and tidy environment surrounding the filament spool **400**. Meanwhile, the supporting shaft assembly **310** can reduce friction produced during rotation of the filament spool, so as to facilitate loading and unloading. Also, since the supporting shaft assembly **310** is supported on the rim **410** of the filament spool **400**, the filament spool **400** can be mounted to the filament replacement mechanism simply by gravity, without other complicated mounting structures. After the filament is consumed, the filament spool **400** can be conveniently taken down from the filament replacement mechanism without disassembling other components, which facilitates replacement.

(30) In some embodiments, referring to FIG. 4, the supporting shaft assembly **310** includes a first supporting shaft set **311** and a second supporting shaft set **312** rotatably connected to the support **100** separately, where the first supporting shaft set **311** and the second supporting shaft set **312** are spaced apart from each other in a circumferential direction of the corresponding filament spool **400**, and a first axis of rotation of the first supporting shaft set **311**, a second axis of rotation of the second supporting shaft set **312**, and an axis of rotation of the corresponding filament spool **400** are parallel to one another.

(31) Referring to FIG. 1, with one side of the filament spool **400** close to the loading and unloading mechanism **320** as a front side, the first supporting shaft set **311** and the second supporting shaft set **312** are mounted at the front side and the rear side of the filament spool **400**, respectively, thereby playing a role of stable support.

(32) FIG. 4 is a stereoscopic structure diagram of a filament replacement mechanism after the removal of a filament spool and a loading and unloading mechanism according to some embodiments of the present disclosure. FIG. 5 is a stereoscopic structure diagram of a filament replacement mechanism after the removal of a filament spool and a top cover according to some embodiments of the present disclosure. Referring to FIG. 4 and FIG. 5, in some embodiments, the first supporting shaft set **311** includes two first supporting shafts **3111** arranged along the first axis

of rotation such that circumferential surfaces of the two first supporting shafts **3111** are in contact with two rims **410** of the corresponding filament spool **400**, respectively, when the corresponding filament spool **400** is mounted in the filament replacement mechanism.

(33) One end of one first supporting shaft **3111** facing away from the other first supporting shaft may be connected to the support **100** by a bearing. For example, a support plate may protrude from the support **100**, an outer race of a bearing may be fixedly connected to the support plate, and an inner race of the bearing may be fixedly connected to the first supporting shaft **3111**, such that the first supporting shaft **3111** may be rotatably connected to the support **100**. Two first supporting shafts **3111** being in contact with the two rims **410** respectively are arranged, such that the volume of the first supporting shaft set **311** can be reduced while stable supporting is achieved, and mounting of other structures is facilitated.

(34) In some embodiments, at least one first supporting shaft **3111** of the two first supporting shafts **3111** is provided with a gear **3112** which is configured to be driven by the loading and unloading mechanism **320** when the 3D printer performs the unloading operation, such that the first supporting shaft **3111** drives the corresponding filament spool **400** to rotate, so as to rewind the filament around the corresponding filament spool **400**.

(35) The gear **3112** may be coaxially connected to a first supporting shaft **3111**, such that the two may rotate synchronously. The loading and unloading mechanism **320** can be in transmission connection with the gear **3112**, for example, the unloading motor can be in transmission connection with the gear **3112** through a transmission mechanism, such that when unloading is performed, the gear **3112** is driven to rotate, and then the filament spool **400** is driven to rotate in the reverse direction under friction force between the first supporting shaft **3111** and the filament spool **400**. During loading and printing, the unloading motor may be in drive disengagement from the gear **3112**, such that the first supporting shaft **3111** may freely rotate following the filament spool **400**.

(36) In some embodiments, one end of the corresponding secondary filament guide tube **210** away from the filament guide opening **220** is arranged adjacent to a middle portion of the first supporting shaft set **311** along the first axis of rotation. This allows the free end of the filament on the filament spool **400** to enter the secondary filament guide tube **210** at a suitable angle relative to the winding portion of the filament spool **400**, which reduces friction produced during loading and unloading. In some embodiments, one end of the corresponding secondary filament guide tube **210** away from the filament guide opening **220** is disposed close to a lower part of a middle position of the two first supporting shafts **3111**, such that the filament can freely fall into the secondary filament guide tube **210** by gravity, and is thus more convenient to use.

(37) In some embodiments, the two first supporting shafts **3111** may be coaxially connected to each other, and the middle portion of the first supporting shaft set **311** is provided with a connecting shaft **3113** having a radial dimension smaller than that of the two first supporting shaft set **311**. The two first supporting shafts **3111** may be connected through the connecting shaft **3113**, which can not only improve the strength of the first supporting shaft set **311**, but also reduce the space of the first supporting shaft set **311**, so as to avoid greater friction produced by the contact between the filament and the supporting shaft set.

(38) In other embodiments, the two first supporting shafts **3111** may also be separated from each other, and the middle portion of the first supporting shaft set **311** is a gap between the two first supporting shafts **3111**, such that the space occupied by the first supporting shaft set **311** may be further reduced.

(39) In some embodiments, the second supporting shaft set **312** includes two second supporting shafts **3121** arranged along the second axis of rotation such that circumferential surfaces of the two second supporting shafts **3121** are in contact with the two rims **410** of the corresponding filament spool **400**, respectively, when a corresponding filament spool **400** is mounted in the filament replacement mechanism. The volume of the first supporting shaft set **311** can be reduced while stable supporting is achieved, and mounting of other structures is facilitated. In some embodiments,

similar to the two first supporting shafts **3111**, the two second supporting shafts **3121** may be either coaxially connected to each other or separated from each other to achieve support of the filament spool **400**.

(40) The second supporting shaft **3121** is disposed in a manner similar to the first supporting shaft **3111**. Referring to FIG. 2, the second supporting shaft **3121** is located at the rear side of the filament spool **400**, the first supporting shaft **3111** is located at the front side of the filament spool **400**, the loading and unloading mechanism **320** and the secondary filament guide tube **210** are disposed at the front side of the filament spool **400**, and such structure is not disposed at the rear side. Reference may be made to the first supporting shaft **3111** for the structure and function of the second supporting shaft **3121**, and details are not described herein.

(41) In some embodiments, the loading and unloading mechanism **320** is disposed on one side of the first supporting shaft set **311** facing away from the second supporting shaft set **312**, and a channel **321** for a filament to penetrate through is further disposed in the loading and unloading mechanism **320**, and is jointed with the corresponding secondary filament guide tube **210**. The channel **321** can play a guiding role, and the dimension of the filament exposed outside the channel **321** and the filament guide unit **200** is reduced, such that the filament is prevented from being wound around other components, and meanwhile, replacement of the filament spool can be facilitated.

(42) One end of the channel **321** facing away from the secondary filament guide tube **210** may be provided with a trumpet-shaped opening, such that the filament can easily enter the channel **321**, and the filament spool **400** can be conveniently replaced. Herein, the term “trumpet-shaped opening” means that one end of the channel **321** facing away from the secondary filament guide tube **210** has a gradually increasing inner diameter in a direction toward the outside of the channel **321**.

(43) Still referring to FIG. 4 to FIG. 5, the support **100** can be provided with an arc-shaped groove **110** extending in a predetermined direction, at least one separator **120** is disposed in the arc-shaped groove and is spaced in the predetermined direction to separate the arc-shaped groove **110** into a plurality of segments, and each of the segments is configured to at least partially accommodate a corresponding filament spool **400** of the plurality of filament spools **400**.

(44) The arc-shaped groove **110** may extend in the predetermined direction such that the surface of the support **100** has a downward recess. The predetermined direction may be an arrangement direction of the plurality of filament spools **400**, and the separator **120** may be of a rib structure. There may be one or a plurality of separators **120**, when there are a plurality of separators **120**, the plurality of separators **120** may be spaced in the predetermined direction to separate the arc-shaped groove **110** into a plurality of segments, and each segment may accommodate at least a portion of a corresponding filament spool **400**, such that the volume of the filament replacement mechanism can be reduced, resulting in a compact structure.

(45) Referring to FIG. 3, the filament guide unit **200** may be located at one side of the support **100** facing away from the arc-shaped groove **110**, that is, the filament spool **400** may be disposed at an upper side of the support **100**, and the filament guide unit **200** may be disposed at a lower side of the support **100**, such that the structure is relatively compact.

(46) Referring to FIG. 4 and FIG. 5, the surface of the support **100** facing the plurality of filament spools **400** is provided with at least one desiccant groove **130** for accommodating a desiccant, and a top cover **140** is detachably mounted at an opening of each of the desiccant grooves **130**, and is provided with a plurality of air holes **150**.

(47) In some embodiments, the desiccant groove **130** may be located at the bottom of the arc-shaped groove **110** and the top cover **140** may be of an arc-shaped structure that fits the groove wall of the arc-shaped groove **110**, that is, the top cover **140** may form part of groove wall of the arc-shaped groove **110** and is in smooth transition with the other groove wall of the arc-shaped groove **110** such that the entire arc-shaped groove **110** may accommodate a corresponding filament spool

400.

(48) A plurality of air holes **150** may be arranged in an array on the top cover **140** to allow the desiccant to absorb moisture from the surrounding environment and keep the surrounding environment dry, thereby providing a suitable working environment for the filament replacement mechanism.

(49) In addition, the desiccant groove **130** may be disposed according to the number of segments. For example, each segment is provided with one desiccant groove **130**. For another example, a plurality of segments may share one desiccant groove **130**, and at this time, the dimension of the separator may be reduced such that the desiccant grooves **130** may span a plurality of segments, and may be provided as needed.

(50) In some embodiments, the filament replacement mechanism further includes a humidity sensor **500** configured to detect the humidity of the surrounding environment of the plurality of filament spools **400**. When the humidity exceeds a threshold value suitable for operation, an alarm can be given to prompt a user to replace the desiccant, or the operation can be stopped, and so on. The humidity sensor **500** may be disposed at various positions, e.g., in a circuit board on the back of the support **100**.

(51) In some embodiments, the filament replacement mechanism further includes a recognition sensor **600** configured to recognize identification information of the filament spool. Information identified by the recognition sensor **600** may be the length of the filament, the color or material of the filament, etc., in the filament spool **400**. The identification information may be, e.g., a bar code or a two-dimensional code provided on the filament spool. The recognition sensor **600** may be configured to scan the identification information and acquire information about the filament spool **400** or the filament. The recognition sensor **600** may be disposed on the support **100**, for example, one recognition sensor **600** may be disposed for each filament spool **400**. Alternatively, a plurality of filament spools **400** may share one recognition sensor **600**.

(52) In some embodiments, the filament replacement mechanism includes a circuit board that may be disposed between two adjacent filament spools **400**, and the recognition sensor **600** may be disposed on the circuit board such that the identification information of the two filament spools **400** can be scanned simultaneously.

(53) FIG. 6 shows a stereoscopic structure diagram of a filament replacement mechanism according to some embodiments of the present disclosure. Referring to FIG. 6, in some embodiments, the filament replacement mechanism further includes: a bottom case **700** and a cover body **800** matched with the bottom case **700**, where the bottom case **700** and the cover body **800** jointly define an accommodating space, the support **100**, the filament guide unit **200** and the plurality of filament replacement units **300** are all accommodated in the accommodating space, and the cover body **800** can move relative to the bottom case **700** to open or close the accommodating space.

(54) In this embodiment, the bottom case **700** may be of a box-type structure, the cover body **800** may be disposed at an opening of the bottom case **700**, and the cover body **800** may be detachably connected to the bottom case **700**, or rotatably connected to the bottom case **700**, such that the accommodating space may be opened or closed.

(55) The bottom case **700** and the cover body **800** may be made of various materials, such as plastic or metal. Through the arrangement of the bottom case **700** and the cover body **800**, the filament replacement mechanism can be transported and used as a whole, which is more conducive to maintaining dryness of the accommodating space, and can also facilitate replacement of the filament spool **400**.

(56) The embodiment of the present disclosure further provides a 3D printing system, including a 3D printer and the filament replacement mechanism according to any one of the embodiments described above. The 3D printer may be a device capable of printing a 3D object, and the structure and function of the filament replacement mechanism are the same as those of the above embodiments, and reference may be made to the above embodiments.

(57) According to the 3D printing system provided by the embodiment of the present disclosure, through the arrangement of the filament guide unit **200** and the filament replacement unit **300** for the filament replacement mechanism, a required filament can be automatically provided for the 3D printer to achieve printing of three-dimensional objects of various colors or various materials, and the unnecessary filament after the return of materials can be automatically wound around the filament spool **400** to keep a clean and tidy environment surrounding the filament spool **400**. Meanwhile, the supporting shaft assembly **310** can reduce friction produced during rotation of the filament spool, so as to facilitate loading and unloading. Also, since the supporting shaft assembly **310** is supported on the rim **410** of the filament spool **400**, the filament spool **400** can be mounted to the filament replacement mechanism simply by gravity, without other complicated mounting structures. After the filament is consumed, the filament spool **400** can be conveniently taken down from the filament replacement mechanism without disassembling other components, which facilitates replacement.

(58) Although the present disclosure is illustrated and described in detail in the drawings and foregoing description, such illustration and description are to be considered illustrative and schematic, rather than restrictive; the present disclosure is not limited to the disclosed embodiments. Variations to the disclosed embodiments can be understood and implemented by those skilled in the art in practicing the claimed subject matter by studying the accompanying drawings, the disclosures, and the appended claims. In the claims, the term “comprising” does not exclude other elements or steps not listed, the indefinite article “a” or “an” does not exclude a plurality, and the term “a plurality” means two or more. The mere fact that certain measures are recorded in dependent claims that are different from one another does not indicate that these measures cannot be combined to achieve a desirable effect.

(59) TABLE-US-00001 Reference numerals: 100: support; 110: arc-shaped groove; 120: separator; 130: desiccant groove; 140: top cover; 150: air hole; 200: filament guide unit; 210: secondary filament guide tube; 220: filament guide opening; 240: multi-channel member; 300: filament replacement unit; 310: supporting shaft assembly; 311: first supporting shaft set; 3111: first supporting shaft; 3112: gear; 3113: connecting shaft; 312: second supporting shaft set; 3121: second supporting shaft; 320: loading and unloading mechanism; 321: channel; 400: filament spool; 410: rim; 500: humidity sensor; 600: recognition sensor; 700: bottom case; 800: cover body.

Claims

1. A filament replacement mechanism for a 3D printer, the filament replacement mechanism comprising: a support; a filament guide unit disposed on the support and comprising a plurality of secondary filament guide tubes and a filament guide opening connecting with the plurality of secondary filament guide tubes, wherein the filament guide opening is configured to be jointed with a main filament guide tube of the 3D printer; and a plurality of filament replacement units disposed on the support, each of the plurality of filament replacement units automatically performing a loading operation and an unloading operation on a corresponding filament spool of a plurality of filament spools for the 3D printer, wherein a surface of the support facing the plurality of filament spools is provided with at least one desiccant groove for accommodating a desiccant, and a top cover is mounted at an opening of each of the desiccant grooves, and is provided with a plurality of air holes; and, wherein the filament guide unit further comprises a multi-channel member connecting the plurality of secondary filament guide tubes to the filament guide opening.
2. The filament replacement mechanism according to claim 1, wherein each of the plurality of filament replacement units comprises: a supporting shaft assembly rotatably connected to the support and configured to be in contact with a rim of the corresponding filament spool so as to support the corresponding filament spool; and a loading and unloading mechanism configured to

drive a filament wound around the corresponding filament spool to enable the filament to be guided into the main filament guide tube via one corresponding secondary filament guide tube of the plurality of secondary filament guide tubes so as to perform a loading operation, and configured to drive the corresponding filament spool to rotate such that the filament withdrawn from the main filament guide tube via the filament guide unit is rewound around the corresponding filament spool so as to perform an unloading operation.

3. The filament replacement mechanism according to claim 2, wherein the supporting shaft assembly comprises a first supporting shaft set and a second supporting shaft set rotatably connected to the support separately and spaced apart from each other in a circumferential direction of the corresponding filament spool, wherein a first axis of rotation of the first supporting shaft set, a second axis of rotation of the second supporting shaft set and an axis of rotation of the corresponding filament spool are parallel to one another.

4. The filament replacement mechanism according to claim 3, wherein the first supporting shaft set comprises two first supporting shafts arranged along the first axis of rotation such that circumferential surfaces of the two first supporting shafts are in contact with two rims of the corresponding filament spool, respectively, when the corresponding filament spool is mounted in the filament replacement mechanism.

5. The filament replacement mechanism according to claim 4, wherein at least one first supporting shaft of the two first supporting shafts is provided with a gear, wherein the gear is configured to be driven by the loading and unloading mechanism to rotate and drive the at least one first supporting shaft to rotate such that the at least one first supporting shaft drives the corresponding filament spool to rotate when the corresponding filament spool is mounted in the filament replacement mechanism, so as to rewind the filament around the corresponding filament spool to perform an unloading operation.

6. The filament replacement mechanism according to claim 4, wherein one end of the corresponding secondary filament guide tube away from the filament guide opening is arranged adjacent to a middle portion of the first supporting shaft set along the first axis of rotation.

7. The filament replacement mechanism according to claim 3, wherein the second supporting shaft set comprises two second supporting shafts arranged along the second axis of rotation such that circumferential surfaces of the two second supporting shafts are in contact with two rims of the corresponding filament spool, respectively, when the corresponding filament spool is mounted in the filament replacement mechanism.

8. The filament replacement mechanism according to claim 3, wherein the loading and unloading mechanism is disposed on one side of the first supporting shaft set facing away from the second supporting shaft set, and is further provided with a channel for the filament to penetrate through, wherein the channel is jointed with the corresponding secondary filament guide tube, and one end of the channel facing away from the corresponding secondary filament guide tube is provided with a trumpet-shaped opening.

9. The filament replacement mechanism according to claim 1, wherein the support is provided with an arc-shaped groove extending in a predetermined direction, and at least one separator is disposed in the arc-shaped groove and is spaced in the predetermined direction to separate the arc-shaped groove into a plurality of segments, wherein each of the segments is configured to at least partially accommodate a corresponding filament spool of the plurality of filament spools.

10. The filament replacement mechanism according to claim 9, wherein the filament guide unit is located on one side of the support facing away from the arc-shaped groove.

11. The filament replacement mechanism according to claim 1, wherein the filament guide unit further comprises a multi-channel member connecting the plurality of secondary filament guide tubes to the filament guide opening.

12. The filament replacement mechanism according to claim 1, further comprising: a humidity sensor configured to detect the humidity of a surrounding environment of the plurality of filament

spools.

13. The filament replacement mechanism according to claim 1, further comprising: a circuit board disposed between every two adjacent filament spools of the plurality of filament spools, wherein a recognition sensor is provided on the circuit board and configured to recognize identification information of the filament spools.

14. The filament replacement mechanism according to claim 1, further comprising: a bottom case; and a cover body matched with the bottom case, wherein the bottom case and the cover body jointly define an accommodating space, the support, the filament guide unit and the plurality of filament replacement units are all accommodated in the accommodating space, and the cover body is movable relative to the bottom case to open or close the accommodating space.

15. A 3D printing system, comprising: a 3D printer; and a filament replacement mechanism comprising: a support; a filament guide unit disposed on the support and comprising a plurality of secondary filament guide tubes and a filament guide opening connecting with the plurality of secondary filament guide tubes, wherein the filament guide opening is configured to be jointed with a main filament guide tube of the 3D printer; and a plurality of filament replacement units disposed on the support, each of the plurality of filament replacement units automatically performing a loading operation and an unloading operation on a corresponding filament spool of a plurality of filament spools for the 3D printer, wherein a surface of the support facing the plurality of filament spools is provided with at least one desiccant groove for accommodating a desiccant, and a top cover is mounted at an opening of each of the desiccant grooves, and is provided with a plurality of air holes; and, wherein the filament guide unit further comprises a multi-channel member connecting the plurality of secondary filament guide tubes to the filament guide opening.

16. The 3D printing system according to claim 15, wherein the support is provided with an arc-shaped groove extending in a predetermined direction, and at least one separator is disposed in the arc-shaped groove and is spaced in the predetermined direction to separate the arc-shaped groove into a plurality of segments, wherein each of the segments is configured to at least partially accommodate a corresponding filament spool of the plurality of filament spools.

17. The 3D printing system according to claim 16, wherein the filament guide unit is located on one side of the support facing away from the arc-shaped groove.

18. The 3D printing system according to claim 15, further comprising: a circuit board disposed between every two adjacent filament spools of the plurality of filament spools, wherein a recognition sensor is provided on the circuit board and configured to recognize identification information of the filament spools.

19. The 3D printing system according to claim 15, further comprising: a bottom case; and a cover body matched with the bottom case, wherein the bottom case and the cover body jointly define an accommodating space, the support, the filament guide unit and the plurality of filament replacement units are all accommodated in the accommodating space, and the cover body is movable relative to the bottom case to open or close the accommodating space.
